

Yash Verma

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EDUCATION

- Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 - Apr 2024
- Relevant Courses:** Data Structures & Algorithms, Operating Systems, System Design, DBMS, Object-Oriented Programming, Distributed Systems.

PROFESSIONAL SUMMARY

- Backend engineer with **3+ years** building scalable distributed systems, microservices, and APIs in production.
- Delivered measurable reliability and performance improvements, including reducing dashboard latency from **3-4 minutes to 3 seconds** through query and data-path optimization.
- Hands-on with **Java/Spring Boot, Ruby on Rails, SQL, Redis, ClickHouse, AWS (RDS), Jenkins, and CI/CD**, with strong ownership of deployments and operational excellence.

EXPERIENCE

- BrowserStack** Mumbai (Mallard West)
SDE - Backend - Ruby, Rails, Express, JavaScript, SpringBoot, Java, ClickHouse, Helm, Jenkins Apr 2025 - Present
 - Improved platform stability by shipping abuse-protection controls that prevented noisy traffic from impacting other users.
 - Led naming standardization across product touchpoints, improving consistency between user-facing and developer-facing workflows.
 - Expanded regional capability for **Australia**, unblocking onboarding and execution for region-specific customer use cases.
 - Contributed to authentication flow simplification for accessibility products, reducing upstream dependency calls and improving response times.
 - Drove a critical performance initiative that reduced dashboard load time from **3-4 minutes to 3 seconds**.
 - Delivered data migration and synchronization improvements that unblocked **3+ teams** and increased adoption of analytics workflows.
 - Standardized connector provisioning through configuration-driven automation, improving reliability and operational consistency.
 - Introduced pre-deployment validation checks across regions, lowering release risk and improving deployment confidence.
- Zus (formerly 0chain)** Cupertino, CA (Remote)
Software Engineer - Blockchain Storage & Mobile - Golang, Kotlin, GitHub Actions, Gin, Grafana Feb 2022 - Mar 2025
 - Architected and deployed two production Android apps: **Bolt** (crypto wallet) and **Vult** (decentralized storage), driving **20%** increase in platform adoption and processing **\$2M+** in transactions.
 - Spearheaded comprehensive **UAT automation suite** for enterprise blockchain storage, reducing critical production bugs by **90%** and cutting release cycles from **2 weeks to 3 days**.
 - Optimized **S3-compatible distributed storage** infrastructure handling **500M+** files and **40TB+** data across **50+ nodes**, achieving consistent sub-**100ms** latency with **99.9%** uptime.
 - Built **real-time monitoring** dashboards using **Grafana/Prometheus**, tracking **blockchain consensus**, storage allocation, and network performance across distributed infrastructure.
 - Developed **automated DevOps workflows** for dynamic **Sharder/Miner** management and view-change protocols, boosting developer productivity by **90%** and reducing manual intervention.
 - Enhanced **multi-wallet concurrency** handling with unified **GoSDK**, supporting **10K+** concurrent transactions and improving load test performance by **300%**.
 - Implemented **Erasur Coding (EC)** optimization for video streaming workloads, reducing storage overhead by **40%** while maintaining data integrity across parity nodes.
- Google Summer of Code Oppia Foundation** Remote
GSOC - Kotlin, GitHub Actions, JUnit, UIAutomator, Gradle, Bazel, AndroidSDK Jun 2021 - Sept 2022
 - Designed robust **CI/CD infrastructure** with automated build triggers for alpha, beta, and test releases.
 - Implemented automated deployment to **Google Drive, GCP Bucket**, and **Google Play Store** tracks.
 - Authored custom **Kotlin scripts** and **GitHub Actions** to optimize CD pipelines for scheduled releases.
- Oppia Foundation Github Link [Link to Merged PRs]** Mountain View, CA (Remote)
Open Source Contributor Android June 2021 - April 2022
 - Contributed to **Gradle to Bazel** migration, achieving **10x** improvement in build speeds.
 - Developed modular **MVVM/MVP** architectures with **tablet support** for multiple screen sizes.
 - Authored **70+ Unit Tests** and **20+ E2E tests** using Espresso, Robolectric, and UIAutomator.
- Gnome Foundation** Remote
Open Source Contributor - Tech Stack: C, XML, GTK+ Sept 2021 - Feb 2022
 - Enhanced **Nautilus file manager** with new path-bar features, improving user efficiency by **20%**.
 - Implemented **drag-and-drop support** and keyboard shortcuts for seamless file organization.

PROJECTS

- Parking Lot Management System:** RESTful API using **Node.js, Express.js, MongoDB** with **Winston** logging and **Jest** testing.
- Airline Ticketing System:** **Spring Boot** MVC architecture with booking, querying, and cancellation features, **Dockerized**.
- Trip Feedback System:** Secure feedback system using **Spring Framework** with user authentication.
- Festify:** Multi-module Android event management app with **Node.js** backend and **MongoDB**.
- ACID-Compliant Database Engine (From Scratch):** Built a lightweight storage engine implementing **transactions, WAL, locking, crash recovery**, and **B-Tree indexing**.
- HTTP Server (From Scratch):** Implemented an **HTTP/1.1** server with request parsing, routing, keep-alive connections, middleware, and concurrent worker handling.

- **TCP Stack Components (From Scratch):** Built socket-level networking modules with connection lifecycle handling, retransmission logic, and flow-control primitives.
- **Event-Driven Processing Pipeline:** Built a Kafka-based workflow with producer services, consumer groups, partitioned topics, schema versioning, retry/backoff, dead-letter queues, and idempotent consumers for reliable event handling.
- **Container Runtime Prototype (Docker-like):** Built an isolated execution environment using Linux **namespaces**, **cgroups**, layered filesystem concepts, and process sandboxing.
- **Distributed Key-Value Store (Raft-based):** Implemented a fault-tolerant KV store with leader election, log replication, and state machine application using the Raft consensus algorithm.
- **Load Balancer (From Scratch):** Developed a TCP/HTTP load balancer with round-robin and least-connections algorithms, health checks, and session persistence features.
- **Distributed File Storage System:** Built a scalable storage solution with sharding, replication, and a custom protocol for client interactions, supporting large file uploads and downloads with fault tolerance.
- **BitTorrent:** Implemented a peer-to-peer file sharing protocol with piece selection, choking/unchoking, and tracker communication for efficient file distribution.

ACHIEVEMENTS

- Practised over **700+** DSA problems on Leetcode.
- Member of **Oppia Foundation** for **2 years**.
- Completed **Cloud Engineering tracks** 30 Days of Google Cloud Program.

TECHNICAL SKILLS

- **Programming & Scripting:** Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Spring Boot, Ruby on Rails, Gin, Express.js, REST APIs, GraphQL, Microservices
- **DevOps & Cloud:** AWS (RDS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, GCP, Linux, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** MySQL, PostgreSQL, MongoDB, Redis, ClickHouse, Query Optimization, JUnit, Mockito, UIAutomator
- **Tools & Libraries:** Git, Apache Kafka, Debezium, CDC, Event-Driven Architecture, Pub/Sub, DLQ, Idempotency, YAML, Retrofit/OkHttp, Gradle, Bazel, Android SDK, System Design